

Question:

John plans to sell all his toys and use the money to buy video games. He has 13 lego sets and he sells them for \$15 each. He ends up buying 8 video games for \$20 each and has \$5 left. How many lego sets does he still have?

Reasoning + Answer:

Let's solve this step by step .

John plans to sell all his toys (specifically , Legos) and use the money to buy video games . The key point in this question is asking :

> "How many Lego sets does he still have?"

But the premise says :

> "John plans to sell all his toys and use the money to buy video games ."

Then it says :

> He has 13 Lego sets and sells them for \$15 each .

So , if he **sells all his Lego sets**, then he would sell 13 of them .

Then it says :

- He sells them for \$15 each $\rightarrow$ total money from sales = $13 \times \$15$
- He uses that money to buy 8 video games at \$20 each $\rightarrow$ total cost = $8 \times \$20$
- He has \$5 left .

Let's verify if that makes sense (by checking money balance):

- 1 . Total money from Legos : $13 \times 15 = \$195$
- 2 . Money spent on video games : $8 \times 20 = \$160$
- 3 . Money left : $\$195 - \$160 = \$35$

But the problem says he has **\$5 left**, not \$35 . So something is wrong .

Ah $\therefore$ therefore , **he did not sell all 13 Legos** . So the statement " he plans to sell all his toys " may be a setup , but the reality $\therefore$ that he ends up with \$5 left $\therefore$ contradicts selling all 13 .

Therefore , the actual number of Lego sets **sold** must be less than 13 , and the rest remain unsold .

So the problem is asking : **How many Lego sets does he still have?**

That would be : initial sets $\rightarrow$ sets sold

Let's define variables :

Let x = number of Lego sets sold

Total money from sales = $15x$

Money spent on video games = $8 \times 20 = \$160$

Money left = \$5

So , total money :

$$15x = 160 + 5 = \$165$$

Now solve for x :

$$15x = 165$$
$$x = 165 / 15 = 11$$

So he sold 11 Lego sets .

He originally had 13 , so :

$$\text{Legos left} = 13 - 11 = 2$$

$\therefore$ Final answer : he still has 2 Lego sets .

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